

Appendix for “Stochastic Analysis of Unstructured DAG-based Distributed Ledger Technologies: Tip Counts, Confirmation Time, and Incoming Approvals”

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Abstract—This appendix provides detailed derivations omitted from the main manuscript because of space limitations. It includes the DTMC transition probabilities, the confirmation-time distribution approximation, and the recursive derivation of the incoming-approval distribution. It also reports the simulation-uncertainty tables and approximation-error summary used to support the mean-value comparisons and the slot-duration sensitivity analysis in the main manuscript.

I. DTMC TRANSITION PROBABILITIES

A. Transition Probabilities

We then derive the transition probability p_{ij} of the DTMC, which represents the probability that the number of tips transitions from i to j in one slot. The state transition depends on two factors: (1) the number of new transactions m arriving in the slot, and (2) the number of existing tips that remain unapproved after the new transactions approve tips.

We assume each of the m new transactions randomly selects K tips for approval. Let $p_{ij}^{(m)}$ denote the conditional transition probability from state i to j given that m new transactions arrive in the slot. The overall transition probability is given by:

$$p_{ij} = \sum_{m=0}^{M_{\max}} p_{ij}^{(m)} \cdot \frac{e^{-\hat{\lambda}} \hat{\lambda}^m}{m!}, \quad \text{for } 1 \leq i, j \leq S_{\max}. \quad (1)$$

There are three possible cases in deriving $p_{ij}^{(m)}$ for a given m :

- **Case 1:** $m = 0$. No new transactions arrive, and no tips are approved. The number of tips remains unchanged, i.e.,

$$p_{ij}^{(0)} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{otherwise.} \end{cases} \quad (2)$$

- **Case 2:** $m > 0$ and $i \leq K$. Each transaction can observe and approve all existing tips. Thus, all tips are guaranteed to be approved, and the only tips remaining in the next slot are the m new ones. Hence,

$$p_{ij}^{(m)} = \begin{cases} 1 & \text{if } j = m, \\ 0 & \text{otherwise.} \end{cases} \quad (3)$$

- **Case 3:** $m > 0$ and $i > K$. Each transaction randomly selects K tips from the i available ones. Let q_i denote the probability that a specific tip is selected by a single transaction:

$$q_i = \frac{K}{i}. \quad (4)$$

Since each transaction selects tips independently, the probability that a given tip is *not* approved by any of the m transactions is:

$$p_i^{(m)} = (1 - q_i)^m. \quad (5)$$

This implies the probability that a tip remains unapproved is $p_i^{(m)}$. Among the i current tips, the number of surviving tips follows a binomial distribution. Letting $j - m$ be the number of such tips (since m new ones are always added), we have:

$$p_{ij}^{(m)} = \binom{i}{j-m} \cdot (p_i^{(m)})^{j-m} \cdot (1 - p_i^{(m)})^{i-(j-m)}. \quad (6)$$

This expression is valid when $j \geq m$ and $j - m \leq i$.

Using these cases, we compute $p_{ij}^{(m)}$ for each (i, j) pair by (2), (3), (6) and integrate over all values of m according to the Poisson distribution in (1).

II. CONFIRMATION-TIME DISTRIBUTION APPROXIMATION

A. Distribution of Confirmation Time

In the previous section, we derived the re-normalized steady-state probability distribution of the tip count in the slotted DAG. Let $\hat{\pi}_i$ denote the probability that there are i tips in steady state after re-normalization. Based on this result, we now estimate the distribution of confirmation time.

Recall that the mean confirmation time was approximated in the previous section by Little’s law as $E[C] = E[T]/\lambda$, where $E[T]$ is the average number of tips and λ is the transaction arrival rate in the original continuous-time DAG. Motivated by this relation, we further construct a discrete approximation of the confirmation time distribution from the steady-state tip-count distribution. This construction should be interpreted as a heuristic approximation rather than an exact distributional derivation. An exact confirmation-time distribution in the original continuous-time DAG would require tracking the joint path of a tagged transaction, the time-varying tip set, all competing tips, and all overlapping PoW completion events until the tagged transaction is first approved. Such a path-dependent joint process is analytically intractable. Therefore, we use the steady-state tip-count distribution as a tractable summary of the random environment seen by a tagged tip and map this distribution to a confirmation-time grid. The approximation error introduced by this mapping is examined in the simulation section of the main manuscript by comparing the analytical CDF with continuous-time DAG simulations.

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Since the proposed analytical model is developed in discrete time while the original DAG evolves in continuous time with arrival rate λ , we approximate the confirmation time by a discrete random variable supported on the grid

$$\left\{ \frac{1}{\lambda}, \frac{2}{\lambda}, \frac{3}{\lambda}, \dots \right\}. \quad (7)$$

Specifically, we map the steady-state probability mass of the tip count to the corresponding confirmation-time grid and approximate

$$\Pr\left(C = \frac{i}{\lambda}\right) \approx \hat{\pi}_i, \quad i = 1, 2, \dots, \lceil \lambda + M_{\max} \rceil. \quad (8)$$

Accordingly, the probability mass function (PMF) of the confirmation time is estimated as

$$p_C\left(\frac{i}{\lambda}\right) = \hat{\pi}_i, \quad i = 1, 2, \dots, \lceil \lambda + M_{\max} \rceil. \quad (9)$$

The cumulative distribution function (CDF) of the confirmation time can then be obtained by summing the PMF:

$$F_C\left(\frac{i}{\lambda}\right) = \Pr\left(C \leq \frac{i}{\lambda}\right) = \sum_{j=1}^i \hat{\pi}_j, \quad i = 1, 2, \dots, \lceil \lambda + M_{\max} \rceil. \quad (10)$$

More generally, for an arbitrary $t \geq 0$, the CDF can be written as

$$F_C(t) = \sum_{j=1}^{\lfloor \lambda t \rfloor} \hat{\pi}_j. \quad (11)$$

It is worth noting that the above approximation is consistent with the mean confirmation time derived previously. In particular,

$$E[C] \approx \sum_{i=1}^{\lceil \lambda + M_{\max} \rceil} \frac{i}{\lambda} \hat{\pi}_i = \frac{1}{\lambda} \sum_{i=1}^{\lceil \lambda + M_{\max} \rceil} i \hat{\pi}_i = \frac{E[T]}{\lambda}. \quad (12)$$

Therefore, the proposed PMF approximation preserves the mean-value relation given by Little's law.

We emphasize that the above formulation is used as an approximation rather than an exact distributional identity. This is because the confirmation dynamics in an unstructured DAG-based DLT do not follow the standard first-in-first-out queueing discipline required by classical distributional forms of Little's law [1]. Nevertheless, as will be shown in the simulation section of the main manuscript, the resulting CDF closely matches that observed in the continuous-time DAG simulation. This indicates that the proposed approximation captures the confirmation-time behavior with good accuracy while remaining analytically simple. The approximation is expected to become more accurate in moderate-to-high load regimes, where the tip process evolves more smoothly and the steady-state tip-count distribution provides a better basis for the confirmation-time behavior.

III. INCOMING-APPROVAL DISTRIBUTION APPROXIMATION

A. Distribution of Incoming Approvals

Consider a tip observed at the beginning of a slot when the system is in state i , i.e., there are i tips in the system. Let N_i denote the total number of incoming approvals eventually received by this tip before it is removed from the tip set. Under the slotted approximation, all transactions arriving in the same

slot observe the same tip set, and approved tips are removed only after the slot ends. Hence, a tip may receive multiple incoming approvals within one slot before departure.

The objective of this subsection is to derive a tractable approximation for the distribution of N_i . The exact distribution would require the joint evolution of the tagged tip's accumulated approvals, its survival event, the approvals received by all other tips, and the future tip-count process. Because these quantities are mutually coupled through the same set of arriving transactions, the exact state description grows combinatorially with the number of tips and is intractable for analysis. We therefore use a recursive heuristic approximation. The derivation proceeds in three steps. First, we derive the one-slot approval distribution of the tagged tip. Second, we derive the state transition conditioned on the tagged tip surviving the current slot, because only surviving tips can continue to receive approvals in later slots. Third, we combine the current-slot approvals and future approvals through a probability-generating-function recursion. This organization separates the random number of approvals received in the current slot from the random future environment faced by the tagged tip.

1) *One-Slot Incoming Approvals*: Let M denote the number of newly arriving transactions in one slot. As in the previous section, M follows a Poisson distribution with mean

$$\hat{\lambda} = \frac{2\beta\lambda}{K}. \quad (13)$$

We first focus on a single slot and ignore future slots. This step answers the following question: given that the tagged tip sees i competing tips at the beginning of the slot, how many of the newly arriving transactions approve it during this slot? For a tip in state i , let A_i denote the number of incoming approvals it receives during the current slot. If $i \leq K$, every new transaction approves all existing tips. Thus, conditioned on $M = m$,

$$A_i \mid (M = m) = m, \quad i \leq K. \quad (14)$$

Since $M \sim \text{Poisson}(\hat{\lambda})$, we obtain

$$A_i \sim \text{Poisson}(\hat{\lambda}), \quad i \leq K. \quad (15)$$

If $i > K$, each new transaction selects K tips uniformly from the i available tips. Hence, the tip is selected by one transaction with probability K/i , and conditioned on $M = m$,

$$A_i \mid (M = m) \sim \text{Binomial}\left(m, \frac{K}{i}\right), \quad i > K. \quad (16)$$

Applying Poisson thinning yields

$$A_i \sim \text{Poisson}\left(\hat{\lambda} \frac{K}{i}\right), \quad i > K. \quad (17)$$

Therefore,

$$A_i \sim \text{Poisson}(\mu_i), \quad (18)$$

where

$$\mu_i = \begin{cases} \hat{\lambda}, & i \leq K, \\ \hat{\lambda} \frac{K}{i}, & i > K. \end{cases} \quad (19)$$

Hence,

$$\Pr(A_i = n) = \frac{e^{-\mu_i} \mu_i^n}{n!}, \quad n = 0, 1, 2, \dots \quad (20)$$

The event $A_i = 0$ is important because it is exactly the event that the tagged tip remains a tip after the current slot. If $A_i \geq 1$,

the tagged tip has received at least one approval and leaves the tip set at the end of the slot; in that case, its total incoming approval count is already determined by the current slot. If $A_i = 0$, the tagged tip survives and the analysis must continue into the next slot. A tip survives to the next slot iff it receives no approval in the current slot. Thus,

$$s_i = \Pr(A_i = 0) = e^{-\mu_i}. \quad (21)$$

2) *Conditional State Transition Given Survival*: Let

$$q_{ij} = \Pr(\text{next state is } j \mid \text{current state is } i, A_i = 0) \quad (22)$$

denote the state transition probability conditioned on survival of the tip.

If $i \leq K$, the tip survives only when no new transaction arrives. In that case, the state remains unchanged, and

$$q_{ij} = \begin{cases} 1, & j = i, \\ 0, & j \neq i, \end{cases} \quad i \leq K. \quad (23)$$

If $i > K$ and the tip survives, none of the m new transactions selects it. Conditioning on survival changes the sampling space for the other tips: since the tagged tip is known not to be selected, the K approvals of each arriving transaction must be chosen from the remaining $i - 1$ tips. This is why the following probability uses $i - 1$ rather than i in the denominator. Conditioned on this event, each new transaction chooses its K approved tips from the remaining $i - 1$ tips. Hence, for any other existing tip, the probability of not being selected by one new transaction is

$$1 - \frac{K}{i - 1}. \quad (24)$$

Conditioned on $M = m$ and survival of the tip, the probability that another existing tip remains unapproved throughout the slot is

$$\alpha_i^{(m)} = \left(1 - \frac{K}{i - 1}\right)^m. \quad (25)$$

For tractability, we approximate the survival events of the other $i - 1$ existing tips as conditionally independent given $M = m$ and survival of the tip. This is the main heuristic step in the incoming-approval derivation. In the exact process, the survival events of different tips are negatively correlated because each arriving transaction selects a limited number of tips without replacement. Keeping all such dependencies would require tracking the full subset of surviving tips and their approval histories. The conditional-independence approximation replaces this high-dimensional dependence structure with a binomial survival model that retains the correct one-tip survival probability. Under this approximation, the number of surviving tips among these $i - 1$ tips is

$$X \sim \text{Binomial}(i - 1, \alpha_i^{(m)}). \quad (26)$$

Since the tip survives and the m new transactions become new tips, the next-state tip count is

$$j = 1 + X + m. \quad (27)$$

Thus, for $i > K$, the conditional transition probability given m is approximated by

$$q_{ij}^{(m)} = \binom{i - 1}{j - m - 1} \left(\alpha_i^{(m)}\right)^{j - m - 1} \left(1 - \alpha_i^{(m)}\right)^{i - j + m}, \quad (28)$$

for all valid j satisfying $0 \leq j - m - 1 \leq i - 1$, and $q_{ij}^{(m)} = 0$ otherwise. Averaging over m gives

$$q_{ij} = \frac{1}{s_i} \sum_{m=0}^{M_{\max}} \Pr(M = m) \Pr(A_i = 0 \mid M = m) q_{ij}^{(m)}, \quad i > K, \quad (29)$$

The factor $1/s_i$ appears because q_{ij} is a conditional transition probability given survival. The summation in the numerator gives the joint probability that (i) m transactions arrive, (ii) the tagged tip receives no approval, and (iii) the next state is j . Dividing by $s_i = \Pr(A_i = 0)$ normalizes this joint probability over all survival events, so that $\sum_j q_{ij} = 1$, where

$$\Pr(M = m) = \frac{e^{-\hat{\lambda}} \hat{\lambda}^m}{m!}, \quad (30)$$

and

$$\Pr(A_i = 0 \mid M = m) = \left(1 - \frac{K}{i}\right)^m. \quad (31)$$

3) *PGF, PMF, CDF, and Mean*: Define the probability generating function (PGF) of N_i as

$$G_i(z) = E[z^{N_i}], \quad |z| \leq 1. \quad (32)$$

We now combine the one-slot approval distribution and the survival-conditioned transition. The PGF recursion is a renewal-type decomposition at the end of the current slot. The tagged tip either departs in the current slot after receiving one or more approvals, or it receives zero approvals and restarts the same problem in the next slot with a new state. In the current slot, the tip has two possibilities. If it receives $n \geq 1$ approvals, it leaves the tip set at the end of the slot and its total incoming approval count is n . If it receives no approval, it survives to the next slot, the system moves to state j with probability q_{ij} , and the future incoming approval count is distributed as N_j . Therefore,

$$G_i(z) = \sum_{n=1}^{\infty} \Pr(A_i = n) z^n + \Pr(A_i = 0) \sum_{j=1}^{\lceil \lambda + M_{\max} \rceil} q_{ij} G_j(z). \quad (33)$$

Since $A_i \sim \text{Poisson}(\mu_i)$,

$$\sum_{n=1}^{\infty} \Pr(A_i = n) z^n = \sum_{n=1}^{\infty} \frac{e^{-\mu_i} \mu_i^n}{n!} z^n = e^{\mu_i(z-1)} - e^{-\mu_i}. \quad (34)$$

Substituting this expression and $\Pr(A_i = 0) = e^{-\mu_i}$ into the recursion yields

$$G_i(z) = e^{\mu_i(z-1)} - e^{-\mu_i} + e^{-\mu_i} \sum_{j=1}^{\lceil \lambda + M_{\max} \rceil} q_{ij} G_j(z). \quad (35)$$

The above system can be solved numerically for all states i . Intuitively, the first two terms, $e^{\mu_i(z-1)} - e^{-\mu_i}$, collect all cases in which the tagged tip receives at least one approval and therefore terminates in the current slot. The last term, $e^{-\mu_i} \sum_j q_{ij} G_j(z)$, collects the survival case: the factor $e^{-\mu_i}$ is the probability of receiving zero approvals, q_{ij} gives the next tip-count state conditioned on that survival, and $G_j(z)$ accounts for all future approvals from the next slot onward.

Since a tip leaves the tip set only after receiving at least one approval, the support of N_i starts from 1. Once $G_i(z)$ is obtained, the PMF of N_i is

$$g_i(n) = \Pr(N_i = n) = [z^n] G_i(z), \quad n = 1, 2, \dots \quad (36)$$

and the corresponding CDF is

$$F_i(n) = \Pr(N_i \leq n) = \sum_{\ell=1}^n g_i(\ell). \quad (37)$$

The mean number of incoming approvals for a tip starting in state i is

$$E[N_i] = G'_i(1). \quad (38)$$

The PGF also provides a direct mean-value check. Since $G_i(1) = 1$, differentiating the recursion at $z = 1$ yields the expected number of approvals accumulated before departure. Differentiating the PGF recursion and evaluating at $z = 1$ gives

$$E[N_i] = \mu_i + e^{-\mu_i} \sum_{j=1}^{\lceil \lambda + M_{\max} \rceil} q_{ij} E[N_j]. \quad (39)$$

Finally, using the re-normalized steady-state tip-count probabilities as an approximation of the initial-state distribution seen by a tip, the overall PGF is approximated by

$$G(z) = \sum_{i=1}^{\lceil \lambda + M_{\max} \rceil} \hat{\pi}_i G_i(z). \quad (40)$$

This averaging step follows the same quasi-stationary viewpoint used for confirmation time: instead of conditioning on the complete history of the DAG at the moment a tagged transaction becomes a tip, we approximate the tagged tip's initial environment by the steady-state distribution of the tip count. The resulting distribution is therefore a steady-state approximation for a representative transaction. Its accuracy is assessed in the simulation section of the main manuscript. Accordingly, the overall PMF and CDF are

$$g(n) = \Pr(N = n) = [z^n] G(z), \quad n = 1, 2, \dots \quad (41)$$

and

$$F(n) = \Pr(N \leq n) = \sum_{\ell=1}^n g(\ell), \quad (42)$$

respectively. The corresponding overall mean incoming approval count is

$$E[N] = \sum_{i=1}^{\lceil \lambda + M_{\max} \rceil} \hat{\pi}_i E[N_i]. \quad (43)$$

It is worth noting that the above distributional framework is consistent with the generalized mean-value result derived earlier. In particular, the overall mean incoming approval count obtained from the PGF-based formulation corresponds to

$$E[N] = E[K_{\text{eff}}] \leq K,$$

where K_{eff} denotes the actual number of approval edges generated by an arriving transaction. When at least K tips are available for every arriving transaction, $K_{\text{eff}} = K$, and the formulation recovers

$$E[N] = K.$$

When fewer than K tips are occasionally available, particularly under light traffic and larger K , the mean incoming approval count can be smaller than K , consistent with the simulation results.

Therefore, the proposed analysis not only characterizes the full distribution of incoming approvals, but also remains consistent with the generalized steady-state mean result derived above. Moreover, this conditional-independence approximation is expected to be more accurate when the number of competing tips is sufficiently large, since the influence of any single tip on the aggregate survival behavior becomes relatively weak in that regime.

IV. SIMULATION UNCERTAINTY TABLES

This section reports the numerical simulation uncertainties referenced in the main manuscript. Each entry is computed from 20 independent runs and reports the sample mean $\pm$ the two-sided 95% confidence interval of the mean using Student's t -distribution.

Appendix Table I summarizes the uncertainty of the main mean-value simulation results. It covers all evaluated arrival rates and K values for the CT-DAG and SL-DAG simulations. The confidence intervals are narrow. The largest half-widths are 0.17 tips, 0.004 time units for confirmation time, and 0.0056 incoming approvals for CT-DAG, and 0.11 tips, 0.002 time units, and 0.0036 incoming approvals for SL-DAG. Thus, the agreement discussed in the main manuscript is not dominated by run-to-run simulation noise.

Appendix Table II reports the uncertainty for the slot-duration sensitivity analysis at $\lambda = 30$. The CT-DAG is the original continuous-time baseline and does not use the slot-duration scaling factor β . Therefore, the CT-DAG values do not change with β . The largest half-widths are 0.114 tips, 0.003 time units, and 0.0003 incoming approvals for CT-DAG. These values show that the observed mean-value robustness to β is statistically stable within the tested settings.

Appendix Table III summarizes the mean-value approximation error between the analytical DTMC predictions and the CT-DAG simulation means for the main comparison setting, i.e., $\beta = 1$, over $\lambda \in \{5, 10, \dots, 50\}$. The table reports the mean absolute error (MAE), mean relative percentage error (Mean RE), maximum relative percentage error (Max RE), and root-mean-square error (RMSE). The largest relative errors occur in the light-traffic regime, especially when $K = 5$ and $\lambda = 5$, where the CT-DAG values are small and the heuristic incoming-approval approximation is most stressed.

REFERENCES

- [1] D. Bertsimas and D. Nakazato, "The distributional little's law and its applications," *Operations Research*, vol. 43, no. 2, pp. 298–310, 1995.

TABLE I

CT-DAG AND SL-DAG SIMULATION UNCERTAINTY FOR THE MEAN-VALUE FIGURES OVER 20 INDEPENDENT RUNS. EACH ENTRY REPORTS THE SAMPLE MEAN $\pm$ THE TWO-SIDED 95% CONFIDENCE INTERVAL OF THE MEAN USING STUDENT'S t -DISTRIBUTION. THE SL-DAG RESULTS USE THE SAME SLOT-DURATION SETTING AS THE MAIN MEAN-VALUE COMPARISONS, I.E., $\beta = 1$.

K	λ	CT-DAG			SL-DAG		
		Tips	Conf. time	Incoming	Tips	Conf. time	Incoming
2	5	6.69 $\pm$ 0.01	1.340 $\pm$ 0.002	1.9968 $\pm$ 0.0002	6.58 $\pm$ 0.01	1.314 $\pm$ 0.002	1.9998 $\pm$ 0.0001
2	10	13.11 $\pm$ 0.04	1.311 $\pm$ 0.003	1.9999 $\pm$ 0.0001	12.83 $\pm$ 0.02	1.283 $\pm$ 0.001	2.0000 $\pm$ 0.0000
2	15	19.54 $\pm$ 0.05	1.303 $\pm$ 0.002	1.9999 $\pm$ 0.0001	19.12 $\pm$ 0.04	1.275 $\pm$ 0.001	2.0000 $\pm$ 0.0000
2	20	25.94 $\pm$ 0.05	1.298 $\pm$ 0.002	1.9999 $\pm$ 0.0000	25.37 $\pm$ 0.05	1.269 $\pm$ 0.001	2.0000 $\pm$ 0.0000
2	25	32.30 $\pm$ 0.10	1.293 $\pm$ 0.004	1.9998 $\pm$ 0.0001	31.67 $\pm$ 0.05	1.266 $\pm$ 0.001	2.0000 $\pm$ 0.0001
2	30	38.81 $\pm$ 0.11	1.293 $\pm$ 0.003	1.9999 $\pm$ 0.0001	37.97 $\pm$ 0.07	1.265 $\pm$ 0.001	1.9998 $\pm$ 0.0000
2	35	45.20 $\pm$ 0.14	1.291 $\pm$ 0.003	1.9998 $\pm$ 0.0001	44.21 $\pm$ 0.09	1.263 $\pm$ 0.001	2.0000 $\pm$ 0.0000
2	40	51.61 $\pm$ 0.17	1.290 $\pm$ 0.003	1.9998 $\pm$ 0.0001	50.54 $\pm$ 0.09	1.262 $\pm$ 0.001	1.9999 $\pm$ 0.0001
2	45	58.08 $\pm$ 0.16	1.290 $\pm$ 0.003	1.9998 $\pm$ 0.0001	56.77 $\pm$ 0.11	1.261 $\pm$ 0.001	1.9998 $\pm$ 0.0001
2	50	64.40 $\pm$ 0.13	1.287 $\pm$ 0.002	1.9997 $\pm$ 0.0001	62.96 $\pm$ 0.10	1.260 $\pm$ 0.001	1.9999 $\pm$ 0.0001
3	5	4.04 $\pm$ 0.01	0.808 $\pm$ 0.002	2.7036 $\pm$ 0.0024	3.77 $\pm$ 0.00	0.754 $\pm$ 0.001	2.7097 $\pm$ 0.0016
3	10	7.23 $\pm$ 0.02	0.723 $\pm$ 0.002	2.9631 $\pm$ 0.0008	7.30 $\pm$ 0.01	0.730 $\pm$ 0.001	2.9956 $\pm$ 0.0003
3	15	10.61 $\pm$ 0.03	0.707 $\pm$ 0.002	2.9963 $\pm$ 0.0002	10.85 $\pm$ 0.01	0.723 $\pm$ 0.000	3.0000 $\pm$ 0.0001
3	20	14.07 $\pm$ 0.04	0.704 $\pm$ 0.002	2.9996 $\pm$ 0.0001	14.38 $\pm$ 0.02	0.719 $\pm$ 0.000	2.9999 $\pm$ 0.0001
3	25	17.50 $\pm$ 0.06	0.700 $\pm$ 0.002	2.9999 $\pm$ 0.0001	17.95 $\pm$ 0.03	0.718 $\pm$ 0.000	2.9999 $\pm$ 0.0001
3	30	21.00 $\pm$ 0.06	0.699 $\pm$ 0.002	2.9998 $\pm$ 0.0001	21.49 $\pm$ 0.03	0.716 $\pm$ 0.000	2.9999 $\pm$ 0.0001
3	35	24.46 $\pm$ 0.06	0.699 $\pm$ 0.001	2.9997 $\pm$ 0.0001	25.01 $\pm$ 0.04	0.715 $\pm$ 0.000	2.9998 $\pm$ 0.0001
3	40	27.92 $\pm$ 0.07	0.697 $\pm$ 0.001	2.9998 $\pm$ 0.0001	28.60 $\pm$ 0.04	0.715 $\pm$ 0.000	2.9998 $\pm$ 0.0001
3	45	31.33 $\pm$ 0.12	0.696 $\pm$ 0.002	2.9997 $\pm$ 0.0001	32.09 $\pm$ 0.05	0.714 $\pm$ 0.000	2.9998 $\pm$ 0.0001
3	50	34.74 $\pm$ 0.10	0.695 $\pm$ 0.002	2.9995 $\pm$ 0.0001	35.68 $\pm$ 0.06	0.713 $\pm$ 0.000	2.9997 $\pm$ 0.0001
4	5	3.47 $\pm$ 0.01	0.694 $\pm$ 0.001	2.9807 $\pm$ 0.0033	2.79 $\pm$ 0.00	0.558 $\pm$ 0.001	2.5988 $\pm$ 0.0026
4	10	5.44 $\pm$ 0.01	0.544 $\pm$ 0.001	3.6081 $\pm$ 0.0032	5.24 $\pm$ 0.01	0.524 $\pm$ 0.000	3.7074 $\pm$ 0.0026
4	15	7.57 $\pm$ 0.02	0.505 $\pm$ 0.001	3.8607 $\pm$ 0.0024	7.79 $\pm$ 0.01	0.520 $\pm$ 0.000	3.9700 $\pm$ 0.0008
4	20	9.81 $\pm$ 0.03	0.491 $\pm$ 0.001	3.9535 $\pm$ 0.0011	10.35 $\pm$ 0.01	0.517 $\pm$ 0.000	3.9981 $\pm$ 0.0002
4	25	12.18 $\pm$ 0.04	0.487 $\pm$ 0.002	3.9861 $\pm$ 0.0006	12.90 $\pm$ 0.02	0.516 $\pm$ 0.000	3.9998 $\pm$ 0.0001
4	30	14.54 $\pm$ 0.05	0.485 $\pm$ 0.001	3.9956 $\pm$ 0.0003	15.46 $\pm$ 0.02	0.515 $\pm$ 0.000	3.9998 $\pm$ 0.0001
4	35	16.89 $\pm$ 0.05	0.483 $\pm$ 0.001	3.9986 $\pm$ 0.0002	17.99 $\pm$ 0.03	0.514 $\pm$ 0.000	3.9997 $\pm$ 0.0001
4	40	19.29 $\pm$ 0.06	0.482 $\pm$ 0.001	3.9994 $\pm$ 0.0002	20.53 $\pm$ 0.04	0.514 $\pm$ 0.000	3.9997 $\pm$ 0.0001
4	45	21.70 $\pm$ 0.08	0.482 $\pm$ 0.002	3.9996 $\pm$ 0.0002	23.05 $\pm$ 0.03	0.513 $\pm$ 0.000	3.9998 $\pm$ 0.0001
4	50	24.12 $\pm$ 0.09	0.483 $\pm$ 0.001	3.9996 $\pm$ 0.0002	25.64 $\pm$ 0.03	0.513 $\pm$ 0.000	3.9997 $\pm$ 0.0002
5	5	3.29 $\pm$ 0.01	0.659 $\pm$ 0.001	3.0971 $\pm$ 0.0039	2.32 $\pm$ 0.00	0.465 $\pm$ 0.000	2.2952 $\pm$ 0.0030
5	10	4.82 $\pm$ 0.01	0.482 $\pm$ 0.001	3.9408 $\pm$ 0.0040	4.14 $\pm$ 0.01	0.414 $\pm$ 0.000	3.7204 $\pm$ 0.0035
5	15	6.32 $\pm$ 0.02	0.422 $\pm$ 0.001	4.4113 $\pm$ 0.0056	6.13 $\pm$ 0.01	0.409 $\pm$ 0.000	4.5766 $\pm$ 0.0036
5	20	7.89 $\pm$ 0.02	0.394 $\pm$ 0.001	4.6781 $\pm$ 0.0031	8.14 $\pm$ 0.01	0.407 $\pm$ 0.000	4.8971 $\pm$ 0.0016
5	25	9.57 $\pm$ 0.03	0.383 $\pm$ 0.001	4.8330 $\pm$ 0.0032	10.15 $\pm$ 0.02	0.406 $\pm$ 0.000	4.9808 $\pm$ 0.0009
5	30	11.26 $\pm$ 0.04	0.375 $\pm$ 0.001	4.9143 $\pm$ 0.0013	12.18 $\pm$ 0.02	0.406 $\pm$ 0.000	4.9973 $\pm$ 0.0003
5	35	13.03 $\pm$ 0.04	0.373 $\pm$ 0.001	4.9572 $\pm$ 0.0015	14.18 $\pm$ 0.02	0.405 $\pm$ 0.000	4.9994 $\pm$ 0.0001
5	40	14.83 $\pm$ 0.04	0.371 $\pm$ 0.001	4.9791 $\pm$ 0.0009	16.20 $\pm$ 0.03	0.405 $\pm$ 0.000	4.9997 $\pm$ 0.0001
5	45	16.63 $\pm$ 0.05	0.370 $\pm$ 0.001	4.9897 $\pm$ 0.0008	18.18 $\pm$ 0.03	0.405 $\pm$ 0.000	4.9998 $\pm$ 0.0001
5	50	18.46 $\pm$ 0.05	0.369 $\pm$ 0.001	4.9954 $\pm$ 0.0005	20.21 $\pm$ 0.03	0.404 $\pm$ 0.000	4.9997 $\pm$ 0.0001

TABLE II

CT-DAG UNCERTAINTY FOR THE SENSITIVITY ANALYSIS FOR FIG. 9 OF THE MAIN MANUSCRIPT AT FIXED $\lambda = 30$. EACH ENTRY REPORTS THE SAMPLE MEAN $\pm$ THE TWO-SIDED 95% CONFIDENCE INTERVAL OF THE MEAN OVER 20 INDEPENDENT RUNS USING STUDENT'S t -DISTRIBUTION. THE CT-DAG BASELINE DOES NOT DEPEND ON β ; THEREFORE, THE SAME CT-DAG VALUES ARE REPEATED ACROSS THE β ROWS FOR READABILITY.

K	Tips	Conf. time	Incoming
2	38.811 $\pm$ 0.114	1.293 $\pm$ 0.003	1.9999 $\pm$ 0.0001
3	21.002 $\pm$ 0.058	0.699 $\pm$ 0.002	2.9998 $\pm$ 0.0001
4	14.535 $\pm$ 0.048	0.485 $\pm$ 0.001	3.9956 $\pm$ 0.0003

TABLE III

APPROXIMATION-ERROR SUMMARY BETWEEN THE ANALYTICAL DTMC PREDICTIONS AND CT-DAG SIMULATION MEANS FOR THE MAIN MEAN-VALUE COMPARISONS AT $\beta = 1$. ERRORS ARE SUMMARIZED OVER $\lambda \in \{5, 10, \dots, 50\}$.

K	Metric	MAE	Mean RE (%)	Max RE (%)	RMSE
2	Tip count	0.715	1.87	2.18	0.827
2	Conf. time	0.024	1.86	2.15	0.025
2	Incoming approvals	0.0005	0.02	0.16	0.001
3	Tip count	0.479	2.69	6.42	0.541
3	Conf. time	0.020	2.72	6.39	0.023
3	Incoming approvals	0.034	1.24	10.96	0.094
4	Tip count	0.854	6.95	19.55	0.962
4	Conf. time	0.038	6.90	19.59	0.050
4	Incoming approvals	0.162	5.04	34.20	0.348
5	Tip count	0.945	10.10	29.48	1.071
5	Conf. time	0.048	10.10	29.47	0.070
5	Incoming approvals	0.420	11.53	61.44	0.723